🌐 Open Standards — Easy KCNA Notes

1. What are Open Standards?

Open Standards are publicly available rules, specifications, or protocols that allow different technologies to work together.

In simple words:

Open Standards = Common rules that everyone can use.

🧠 Easy example

Imagine everyone agrees to use the same language:

Person A — English — Person B

Because both understand the same language, they can communicate.

Similarly, computers and software use common standards to communicate.

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2. Why do we need Open Standards?

Imagine you have:

☁️ Cloud A

☁️ Cloud B

💻 Application A

💻 Application B

If they follow common standards, they can communicate and work together.

Cloud A

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Open Standards

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Cloud B

Without common standards, every company might create its own completely different way of communicating.

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3. Main Benefits ⭐

🔗 Interoperability

Different systems can **work together**.

System A ↔ System B

🔒 Avoid Vendor Lock-in

You are less dependent on one specific vendor.

For example, using widely adopted standards can make it easier to move between technologies.

🌍 Portability

Applications and data can be easier to move between different environments.

🤝 Collaboration

Different organizations and developers can build technologies that follow the same standards.

🔄 Compatibility

Different tools can understand and communicate with each other.

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4. Open Standards vs Proprietary Standards

🌐 Open Standard

Rules/specifications are publicly available and can be implemented by different vendors.

Vendor A

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Open Standard

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Vendor B

🔒 Proprietary

A technology or specification is controlled by a particular company or organization and may have restrictions on its use or implementation.

Company A

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Proprietary Technology

🧠 **Easy memory:**

Open = shared/public rules

Proprietary = controlled by a particular vendor

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5. Examples

Some important technologies/protocols commonly associated with open standards include:

- **HTTP/HTTPS** → Web communication
- **DNS** → Name resolution
- **TCP/IP** → Network communication
- **TLS** → Secure communication
- **OCI specifications** → Container image/runtime standards
- **Kubernetes APIs** → Standardized interfaces for Kubernetes resources

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6. Open Standards in Cloud Native ☁

Open standards are especially important in **Cloud Native** because cloud-native environments use many different tools.

For example:

Developer

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Container

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Kubernetes

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Cloud Provider

Using common standards helps these different technologies communicate and work together.

⭐ **KCNA Revision Notes**

Open Standards

- 🌐 Publicly available rules/specifications.
- 🤝 Allow different technologies to work together.
- 🔗 Provide **interoperability**.
- 🔄 Improve **portability**.
- 🔒 Can help reduce **vendor lock-in**.
- 🤝 Encourage collaboration and compatibility.
- Important in **Cloud Native** environments.

🧠 **One-line definition**

Open standards are publicly available specifications that enable different systems, vendors, and technologies to communicate and interoperate.

⭐ **Easy memory trick**

OPEN = Everyone can follow the common rules. 🌍